Uniform plane wave propagation

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For free space:

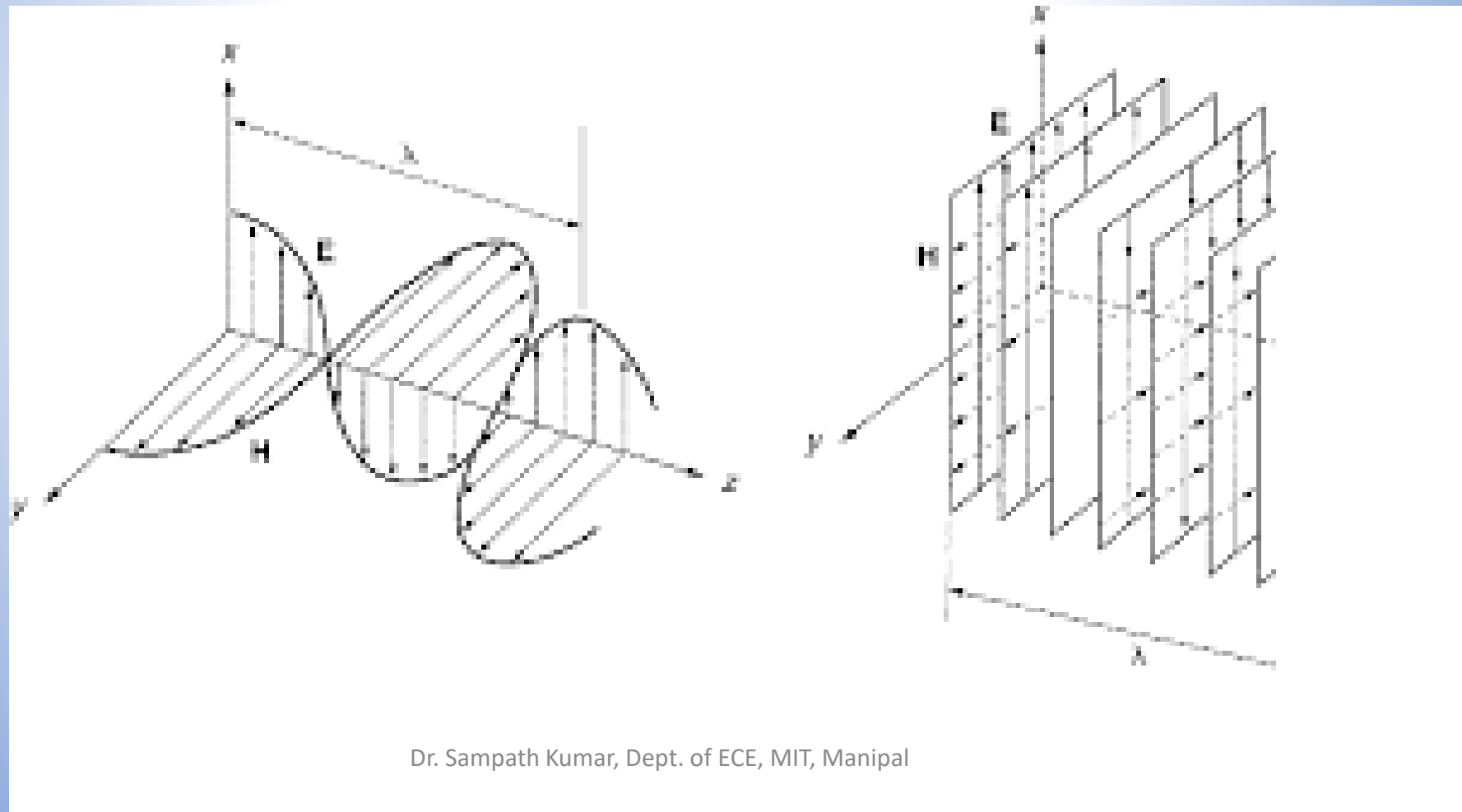
$$\beta = \frac{2\pi}{\lambda} \qquad \beta = \omega\sqrt{\mu_0 \epsilon_0}$$

$$\text{velocity} = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$$

$$\frac{E_0}{H_0} = \eta = \sqrt{\frac{\mu_0}{\epsilon_0}} = 377\ \Omega$$

Uniform plane waves

- Planes of uniform fields propagating in the form of a wave



- Revision: When E and H fields are complex exponentials – Maxwell's equations in phasor form

$$\mathbf{E} = \mathrm{Re}\left\{\tilde{\mathbf{E}} e^{j\omega t}\right\}$$

$$\mathbf{B} = \mathrm{Re}\left\{\tilde{\mathbf{B}} e^{j\omega t}\right\}$$

$$\nabla \times \left[\mathrm{Re}\left\{\tilde{\mathbf{E}} e^{j\omega t}\right\}\right] = -\frac{\partial}{\partial t}\left[\mathrm{Re}\left\{\tilde{\mathbf{B}} e^{j\omega t}\right\}\right]$$

$$\mathrm{Re}\{[\nabla \times \tilde{\mathbf{E}}] e^{j\omega t}\} = -\mathrm{Re}\left\{\tilde{\mathbf{B}} \frac{\partial}{\partial t} e^{j\omega t}\right\}$$

$$= -\mathrm{Re}\left\{\tilde{\mathbf{B}}\, j\omega e^{j\omega t}\right\}$$

$$= \mathrm{Re}\left\{[-j\omega \tilde{\mathbf{B}}] e^{j\omega t}\right\}$$

$$\boxed{\nabla \times \tilde{\mathbf{E}} = -j\omega \tilde{\mathbf{B}}} = -\mu j\omega H$$

$$\boxed{\nabla \times \tilde{\mathbf{H}} = \tilde{\mathbf{J}} + j\omega \tilde{\mathbf{D}}}$$

$$= \sigma E + j\omega \epsilon E$$

$$\nabla \times \vec{H} = (\sigma + j\omega\epsilon)\vec{E}$$

Note: other 2 equations (Gauss's law) unchanged //

Consider $\nabla \times \vec{E} = -j\omega\mu \vec{H}$

Take curl on both sides

$$\nabla \times \nabla \times \vec{E} = -j\omega\mu (\nabla \times \vec{H})$$

But $\nabla \times \vec{H} = (\sigma + j\omega\epsilon)\vec{E}$

$$\therefore \nabla \times \nabla \times \vec{E} = -j\omega\mu(\sigma + j\omega\epsilon)\vec{E}$$

Use vector identity $\boxed{\nabla \times \nabla \times \vec{E} = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E}}$

∇^2 → Laplacian operator

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial}{\partial z^2}\right)\vec{E}$$

If no stored charges $\nabla \cdot D = \rho_v = 0$

Note: Movement of charge = current may be present

$$\therefore \quad \nabla \times \nabla \times \vec{E} = -\nabla^2 \vec{E}$$

$$\Rightarrow \quad -\nabla^2 \vec{E} = -\mu j\omega (\sigma + j\omega\epsilon)\vec{E}$$

Similarly $-\nabla^2 H = -\mu j\omega(\sigma + j\omega\epsilon)\vec{H}$ (Using the Curl on othe Maxwell's eqn)

Negative sign can be removed

On simplification $\Rightarrow \nabla^2 \vec{E} = -\omega^2 \mu \left(\epsilon - j\frac{\sigma}{\omega}\right)\vec{E}$

write it as Complex Permitivity $\epsilon' - j\epsilon''$

$\epsilon' = \epsilon_0 \epsilon_r$

$\epsilon'' = \frac{\sigma}{\omega}$

$$\nabla^2 \vec{E} = -\omega^2 \mu (\epsilon' - j\epsilon'')\vec{E}$$

In compact form $\nabla^2 \vec{E} = -k^2 \vec{E}$

where $jk = \sqrt{-\omega^2 \mu (\epsilon' - j\epsilon'')}$

$$\nabla^2 \vec{E} = -k^2 \vec{E}$$

Similarly we can derive $\nabla^2 \vec{H} = -k^2 \vec{H}$ } vector wave equations

$$jk = \sqrt{-\omega^2 \mu (\epsilon' - j\epsilon'')}$$

K, α, β

Using the calculus one can write it as $jk = \alpha + j\beta$

where

$$\alpha = \sqrt{\frac{\omega^2 \mu \epsilon'}{2}\left[\left(1+\left(\frac{\epsilon''}{\epsilon'}\right)^2\right)^{\frac{1}{2}} - 1\right]}$$

and

$$\beta = \sqrt{\frac{\omega^2 \mu \epsilon'}{2}\left[\left(1+\left(\frac{\epsilon''}{\epsilon'}\right)^2\right)^{\frac{1}{2}} + 1\right]}$$

$$\epsilon' = \epsilon_0 \epsilon_r$$

$$\epsilon'' = \frac{\sigma}{\omega}$$

Note: Study taking root of a complex number

Formula

$$\sqrt{a+jb} = \sqrt{\frac{a+\sqrt{a^2+b^2}}{2}} + j\sqrt{\frac{-a+\sqrt{a^2+b^2}}{2}}$$


Thank
you